

$$\Rightarrow 6x = 12$$

$$\Rightarrow x = 2.$$

Difference in their ages = $(7x - 3x)$ years
 $= 4x$ years = (4×2) years = 8 years.

18. Let Sulekha's age be $9x$ years. Then, Arunima's age = $8x$ years.

$$\therefore \frac{9x+5}{8x+5} = \frac{10}{9} \Rightarrow 9(9x+5) = 10(8x+5)$$

$$\Rightarrow 81x + 45 = 80x + 50$$

$$\Rightarrow x = 5.$$

Difference in their ages = $(9x - 8x)$ years
 $= x$ years = 5 years.

19. Let A's age be $5x$ years. Then, B's age = $6x$ years.

$$\therefore \frac{5x+6}{6x+6} = \frac{6}{7} \Rightarrow 7(5x+6) = 6(6x+6)$$

$$\Rightarrow 35x + 42 = 36x + 36$$

$$\Rightarrow x = 6.$$

B's age 5 years ago = $(6x - 5)$ years
 $= (6 \times 6 - 5)$ years = 31 years.

20. Let daughter's age be x years. Then, mother's age = $3x$ years.

$$(3x + 12) = 2(x + 12)$$

$$\Rightarrow 3x + 12 = 2x + 24$$

$$\Rightarrow x = 12.$$

$\therefore$ Daughter's age today = 12 years.

21. Let daughter's age be x years. Then, mother's age = $(56 - x)$ years.

$$(56 - x) + 4 = 3(x + 4)$$

$$\Rightarrow 60 - x = 3x + 12$$

$$\Rightarrow 4x = 48$$

$$\Rightarrow x = 12.$$

$\therefore$ Daughter's age = 12 years, Mother's age = 44 years.

22. Let mother's age be $2x$ years. Then, son's age = x years.

$$(2x - 10) = 3(x - 10)$$

$$\Rightarrow 2x - 10 = 3x - 30$$

$$\Rightarrow x = 20.$$

Son's age = 20 years.

23. Ram's father's age = 50 years, Ram's age

$$= \left(\frac{3}{5} \times 50\right) \text{ years} = 30 \text{ years.}$$

$$\text{Ram's wife's age} = \left(\frac{4}{5} \times 30\right) \text{ years} = 24 \text{ years.}$$

$$\text{Ram's son's age} = \left(\frac{1}{3} \times 24\right) \text{ years} = 8 \text{ years.}$$

24. Let Subhash's age be $3x$ years. Then, Prasad's age = $6x$ years and Amar's age = $7x$ years.

$$\therefore 7x - 6x = 10$$

$$\Rightarrow x = 10.$$

Required difference = $(6x - 3x)$ years
 $= 3x$ years = (3×10) years = 30 years.

25. Let Rajan's age 8 years ago be x years. His present age = $(x + 8)$ years.

$$\therefore x + 8 = \frac{6}{5}x \Rightarrow 5x + 40 = 6x \Rightarrow x = 40.$$

Rajan's sister's age 8 years ago = $(40 - 10)$ years = 30 years.

His sister's age now = $(30 + 8)$ years = 38 years.

26. Let their present ages be x years and $(x - 20)$ years.

$$(x - 5) = 5[(x - 20) - 5]$$

$$\Rightarrow (x - 5) = 5(x - 25)$$

$$\Rightarrow (x - 5) = 5x - 125$$

$$\Rightarrow 4x = 120$$

$$\Rightarrow x = 30.$$

$\therefore$ Their present ages are 30 years and 10 years.

27. M-Mother, F-Father, S-Son and D-Daughter.

$$F = 4S, D = \frac{1}{3}M, M = F - 6 \text{ and } S = D - 3$$

$$\therefore M = 3D = 3(S + 3)$$

$$= 3S + 9 = \frac{3}{4}F + 9 = \frac{3}{4}(M + 6) + 9$$

$$= \frac{3}{4}M + \frac{3}{4} \times 6 + 9$$

$$\Rightarrow \left(M - \frac{3}{4}M\right) = \left(\frac{9}{2} + 9\right) \Rightarrow \frac{1}{4}M = \frac{27}{2}$$

$$\Rightarrow M = \left(\frac{27}{2} \times 4\right) = 54 \text{ years.}$$

$\therefore$ The mother is 54 years old.

28. $A : B : C = 4 : 7 : 9$ and $(A + B + C) = 56 + (8 + 8 + 8) = 80.$

$$\therefore C's \text{ age} = \left(80 \times \frac{9}{20}\right) \text{ years} = 36 \text{ years.}$$

29. Let B's age be x years. Then, A's age = $(x + 9)$ years.

$$(x + 9) + 10 = 2(x - 10)$$

$$\Rightarrow x + 19 = 2x - 20$$

$$\Rightarrow x = 39.$$

B's age = 39 years.

30. Mother's age when Reenu's brother was born = 36 years.

Father's age when Reenu's brother was born = $(38 + 4)$ years = 42 years.

Required difference = $(42 - 36)$ years = 6 years.

31. Let the ages of children be $x, (x + 3), (x + 6), (x + 9)$ and $(x + 12)$ years.

$$\text{Then, } x + x + 3 + x + 6 + x + 9 + x + 12 = 50$$

$$\Rightarrow 5x = 20 \Rightarrow x = 4.$$

$\therefore$ Age of youngest child = 4 years.

32. Let the present age of the man be x years. Then

$$3(x + 3) - 3(x - 3) = x$$

$$\Rightarrow (3x + 9) - (3x - 9) = x$$

$$\Rightarrow x = 18.$$

$\therefore$ The present age of the man is 18 years.

33. Let their ages 10 years ago be $2x$ years, $3x$ years and $4x$ years respectively.

$$\text{Then, } (2x + 10) + (3x + 10) + (4x + 10) = 93$$

$$\Rightarrow 9x + 30 = 93$$